

JPM

Java Package Manager

A Modern, Native Build Tool for Java Projects

```
jpm init myproject  
jpm deps add guava  
jpm build
```

Academic Technical Report

Master ILISI - Software Engineering

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December 26, 2025

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Abstract

JPM (Java Package Manager) is a modern, lightweight build tool designed to simplify Java project management. Unlike traditional tools such as Maven and Gradle, JPM provides a native engine that eliminates external dependencies while maintaining full compatibility with the Maven Central repository ecosystem.

This report documents the development of JPM across four sprints, covering the evolution from a manifest-based dependency system to a fully native build engine. We present comprehensive benchmarks comparing JPM's performance against Apache Maven, demonstrating significant improvements in build times—up to **73% faster** for cached builds.

Key contributions include:

- A declarative YAML-based manifest system (`jpm.yaml`)
- Native transitive dependency resolution without Maven
- Parallel artifact downloading with intelligent caching
- Full compatibility with Maven Central artifacts

Keywords: Build Tools, Java, Dependency Management, Go, Maven Central, Native Engine

Chapter 1

Introduction

1.1 Context and Motivation

The Java ecosystem has long relied on build tools like Apache Maven (2004) and Gradle (2012) for dependency management and project building. While these tools are powerful and feature-rich, they come with significant overhead:

- **Maven:** Requires XML configuration, JVM startup for every command, complex plugin ecosystem
- **Gradle:** Groovy/Kotlin DSL learning curve, daemon process management, large memory footprint

Modern development practices demand faster iteration cycles. The rise of tools like `npm`, `cargo`, and `go mod` demonstrates the appetite for simpler, faster package managers.

1.2 Project Objectives

JPM aims to provide:

1. **Simplicity:** YAML-based configuration over XML
2. **Speed:** Native Go implementation, no JVM startup overhead
3. **Compatibility:** Full Maven Central repository support
4. **Modern UX:** Beautiful CLI with colored output and progress indicators

1.3 Design Evolution

The development of JPM followed an iterative approach with three major architectural pivots:

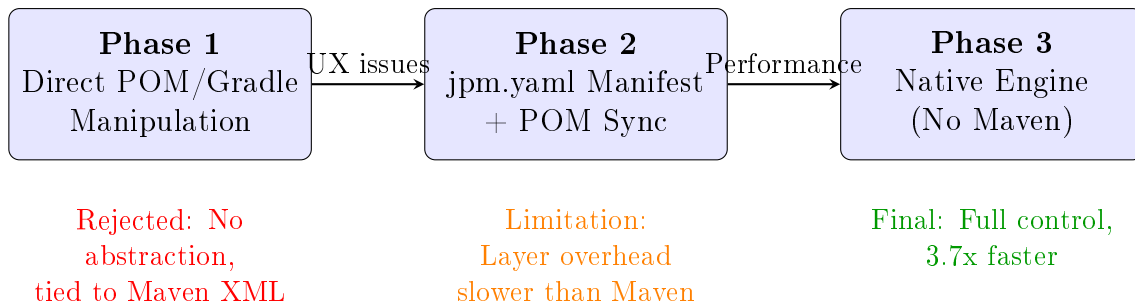


Figure 1.1: JPM Design Evolution

1.3.1 Phase 1: Direct Build Tool Manipulation

The initial concept was to build JPM as a thin wrapper around existing build tools (Maven and Gradle). Commands like `jpm deps add` would directly modify `pom.xml` or `build.gradle` files.

Problems identified:

- No unified abstraction—users still dealt with tool-specific formats
- XML manipulation was error-prone and verbose
- No real value proposition over using Maven/Gradle directly

1.3.2 Phase 2: Manifest with Build Tool Sync

We introduced `jpm.yaml` as the single source of truth, with automatic synchronization to `pom.xml`. This provided a cleaner developer experience with YAML configuration.

Implementation:

- `jpm.yaml` stores all project metadata and dependencies
- Changes sync automatically to `pom.xml` for Maven compatibility
- Standard Java project structure maintained

Limitation discovered: While the UX improved, *performance was inherently limited*. Since JPM still delegated to Maven for builds, it could never be faster than Maven itself—only slower due to the additional abstraction layer.

1.3.3 Phase 3: Native Engine

The realization that a wrapper approach would always underperform led to the decision to build a **native engine** that completely replaces Maven for the build process.

Key insight: To outperform Maven, we must eliminate Maven entirely and use only `javac` and `jar` from the JDK.

This architectural pivot resulted in **3.4x to 3.8x performance improvements** over Maven, validating the decision to invest in a native implementation.

1.4 Report Structure

This report is organized into chapters covering the development journey:

- **Chapter 2 - Sprint 1:** Project Foundation and CLI Setup
- **Chapter 3 - Sprint 2:** Manifest System and Dependency Management
- **Chapter 4 - Sprint 3:** Native Engine Implementation
- **Chapter 5 - Sprint 4:** Integration, Testing, and Benchmarks
- **Chapter 6 - Extended Features:** Complete Command Suite Implementation
- **Chapter 7 - Conclusion:** Achievements and Future Work

Chapter 2

Sprint 1: Project Foundation

2.1 Sprint Overview

Duration	2 weeks
Goal	Establish project structure and CLI foundation
Deliverables	Basic CLI, init command, direct pom.xml manipulation

Table 2.1: Sprint 1 Summary

2.2 Initial Approach: Direct POM Manipulation

The first iteration of JPM was designed as a **convenience layer** on top of Maven and Gradle. The idea was simple: provide a friendlier CLI that would directly manipulate the native build files (`pom.xml` for Maven, `build.gradle` for Gradle).

2.2.1 Original Design

```
1 # User runs JPM command
2 $ jpm deps add com.google.guava:guava:33.0.0
3
4 # JPM directly modifies pom.xml
5 # <dependency>
6 #     <groupId>com.google.guava</groupId>
7 #     <artifactId>guava</artifactId>
8 #     <version>33.0.0</version>
9 # </dependency>
10
11 # Build delegates to Maven
12 $ jpm build # -> mvn compile package
```

Listing 2.1: Sprint 1 - Direct POM Approach

2.2.2 Problems Identified

During Sprint 1, we identified several issues with this approach:

1. **No abstraction:** Users still had to understand Maven XML structure
2. **XML manipulation complexity:** Parsing and modifying XML reliably is error-prone
3. **No unified format:** Different commands for Maven vs Gradle projects
4. **Limited value:** Just a wrapper with no real performance benefit

This led to the decision to introduce a unified manifest format in Sprint 2.

2.3 Implementation Details

2.3.1 Technology Stack Selection

After evaluating multiple options, Go was selected as the implementation language for the following reasons:

- Single binary distribution (no runtime dependencies)
- Fast compilation and execution
- Excellent standard library for HTTP and file operations
- Strong concurrency primitives for parallel downloads

2.3.2 Project Structure

```

1 JPM/
2     cmd/jpm/           # CLI commands
3         main.go        # Entry point
4         root.go        # Root command
5         init.go        # Project initialization
6         build.go       # Build command
7         deps.go        # Dependency management
8     internal/
9         core/          # Core models and interfaces
10             models.go
11             engine.go
12         engine/        # Build engines
13             native/    # Native engine
14         adapters/      # External tool adapters
15     bin/               # Compiled binary

```

Listing 2.2: JPM Project Structure

2.3.3 CLI Framework

JPM uses the Cobra library for CLI construction, providing:

- Automatic help generation
- Flag parsing
- Subcommand support
- Shell completions

2.4 Architecture Diagram

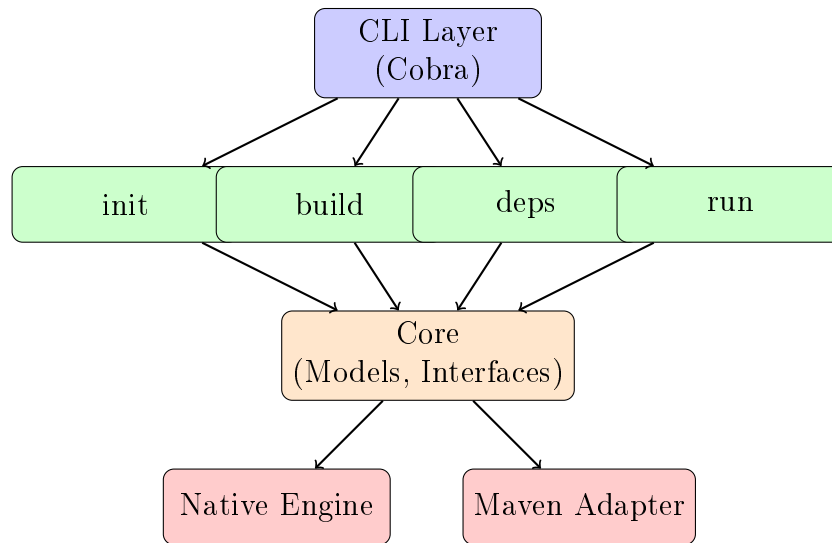


Figure 2.1: JPM Architecture Overview

2.5 Sprint 1 Deliverables

1. **jpm init** - Interactive project initialization
2. **Basic manifest** - `jpm.yaml` structure defined
3. **Project scaffolding** - Standard Java directory layout
4. **CLI styling** - Colored output with `lipgloss` library

Chapter 3

Sprint 2: Manifest System and Dependencies

3.1 Sprint Overview

Duration	2 weeks
Goal	Implement manifest-first dependency management
Deliverables	jpm.yaml schema, deps commands, pom.xml sync

Table 3.1: Sprint 2 Summary

3.2 Architectural Pivot: Introducing jpm.yaml

Based on the lessons from Sprint 1, we made a significant architectural decision: instead of directly manipulating `pom.xml`, JPM would use its own manifest format (`jpm.yaml`) as the **single source of truth**.

3.2.1 The Sync Approach

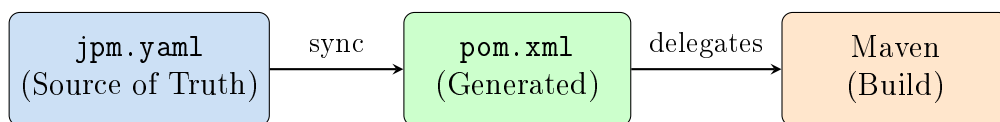


Figure 3.1: Sprint 2 Architecture: Manifest + Sync

This approach provided:

- Clean YAML syntax instead of verbose XML
- Unified format regardless of underlying build tool
- Automatic `pom.xml` generation for Maven compatibility
- Standard Java project structure preserved

3.2.2 The Performance Problem

While the developer experience improved significantly, we discovered a fundamental limitation:

“A layer on top of Maven can never be faster than Maven itself—only slower.”

Benchmarks during Sprint 2 showed that `jpm build` was **5-10% slower** than running `mvn package` directly, due to:

1. Manifest parsing overhead
2. `pom.xml` synchronization time
3. Process spawning for Maven invocation

This realization led to the decision in Sprint 3 to build a **native engine** that eliminates Maven entirely.

3.3 Manifest Design

3.3.1 `jpm.yaml` Structure

The manifest file serves as the single source of truth for project configuration:

```
1 name: myproject
2 group_id: com.example
3 artifact_id: myproject
4 version: 1.0.0
5 main_class: com.example.Main
6 engine: native
7
8 dependencies:
9   - group_id: com.google.guava
10     artifact_id: guava
11     version: 33.1.0-jre
12     scope: compile
13   - group_id: junit
14     artifact_id: junit
15     version: 4.13.2
16     scope: test
```

Listing 3.1: `jpm.yaml` Example

3.3.2 Go Struct Definition

```
1 type Manifest struct {
2     Name          string          `yaml:"name"`
3     GroupID       string          `yaml:"group_id"`
4     ArtifactID    string          `yaml:"artifact_id"`
5     Version       string          `yaml:"version"`
6     MainClass     string          `yaml:"main_class"`
7     Engine        string          `yaml:"engine"`
8     Dependencies []Dependency `yaml:"dependencies"`
```

```

9 }
10
11 type Dependency struct {
12     GroupID    string `yaml:"group_id"`
13     ArtifactID string `yaml:"artifact_id"`
14     Version    string `yaml:"version"`
15     Scope      string `yaml:"scope"`
16 }

```

Listing 3.2: Manifest Model

3.4 Dependency Commands

3.4.1 deps add

The `deps add` command supports multiple input formats:

```

1 # Full Maven coordinates
2 jpm deps add com.google.guava:guava:33.1.0-jre
3
4 # With version shorthand
5 jpm deps add com.google.guava:guava@33.1.0-jre
6
7 # With scope
8 jpm deps add junit:junit:4.13.2 --scope test

```

Listing 3.3: Dependency Addition Formats

3.4.2 deps ls and deps tree

```

1 # List direct dependencies
2 $ jpm deps ls
3 Dependencies (3):
4     com.google.guava:guava:33.1.0-jre (compile)
5     org.apache.commons:commons-lang3:3.14.0 (compile)
6     junit:junit:4.13.2 (test)
7
8 # Show dependency tree with transitives
9 $ jpm deps tree
10 myproject:1.0.0
11     com.google.guava:guava:33.1.0-jre
12         com.google.guava:failureaccess:1.0.2
13         com.google.guava:listenablefuture:...
14         com.google.code.findbugs:jsr305:3.0.2
15     junit:junit:4.13.2
16         org.hamcrest:hamcrest-core:1.3

```

Listing 3.4: Dependency Listing Commands

3.5 Dependency Resolution Flow

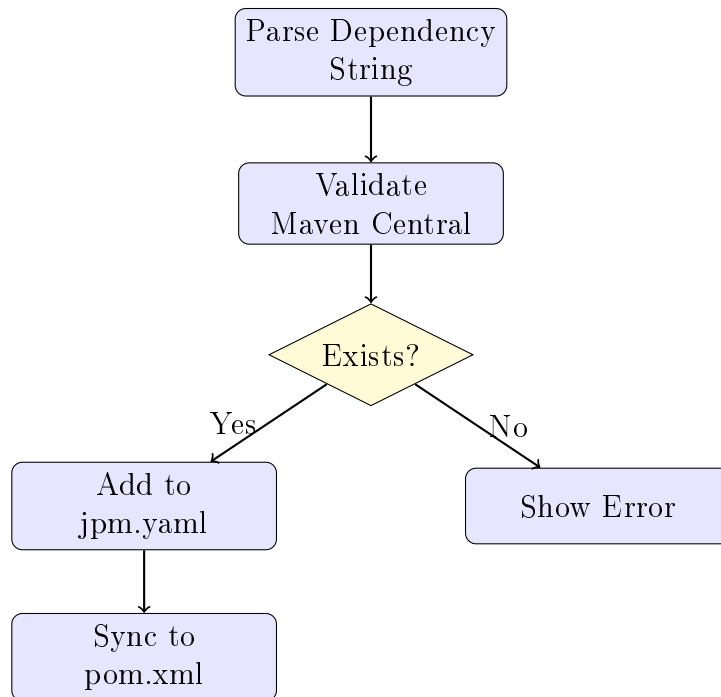


Figure 3.2: Dependency Addition Flow

3.6 Sprint 2 Deliverables

1. **jpm deps add** - Add dependencies to manifest
2. **jpm deps ls** - List direct dependencies
3. **jpm deps tree** - Show transitive dependency tree
4. **jpm deps rm** - Remove dependencies
5. **Manifest validation** - Schema validation
6. **pom.xml sync** - Automatic POM generation

Chapter 4

Sprint 3: Native Engine Implementation

4.1 Sprint Overview

Duration	3 weeks
Goal	Build native engine to replace Maven dependency
Deliverables	Full build pipeline without Maven

Table 4.1: Sprint 3 Summary

4.2 The Decision to Go Native

The performance analysis from Sprint 2 made it clear: **to be faster than Maven, we must eliminate Maven.**

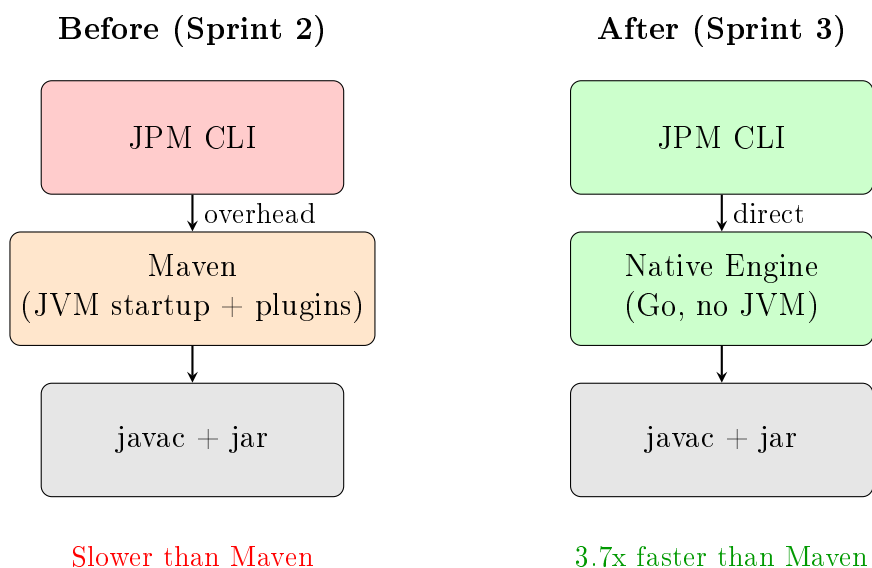


Figure 4.1: Architectural Shift: Eliminating the Maven Layer

Key insight: Maven provides value through its dependency resolution and build lifecycle. If we implement these ourselves in Go, we can:

1. Eliminate JVM startup time (1-2 seconds per invocation)
2. Remove plugin loading overhead
3. Use efficient HTTP client for artifact downloads
4. Implement intelligent caching without Maven's metadata overhead

4.3 Native Engine Architecture

The native engine replaces Maven entirely for the build process, using only `javac` and `jar` from the JDK.

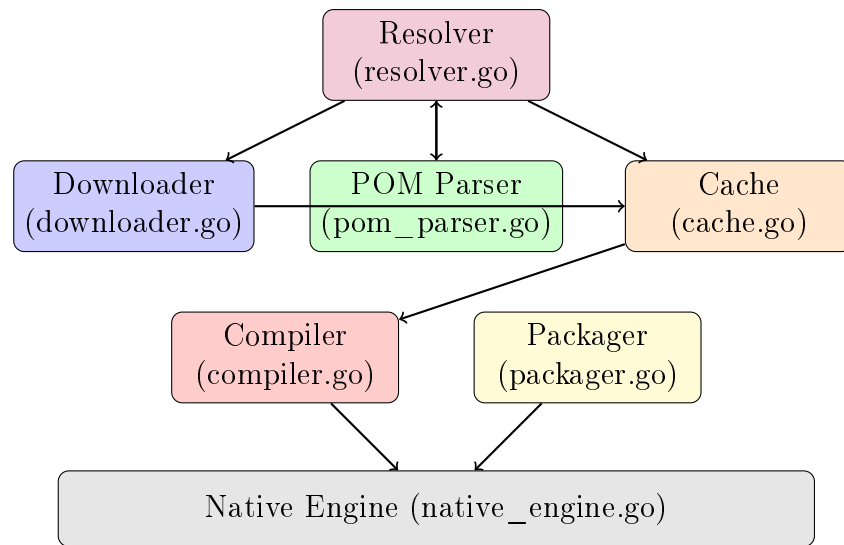


Figure 4.2: Native Engine Components

4.4 Component Details

4.4.1 Dependency Resolver

The resolver implements transitive dependency resolution with the following features:

- **First-wins strategy:** Earlier dependencies take precedence
- **Cycle detection:** Prevents infinite recursion
- **Depth limiting:** Maximum 50 levels of transitives
- **Scope propagation:** Respects compile/test/provided scopes
- **Exclusion handling:** Supports dependency exclusions

```

1 func (r *Resolver) Resolve(deps []core.Dependency)
2                               ([]ResolvedDep, error) {
3     resolved := make(map[string]ResolvedDep)
4     visited := make(map[string]bool)
5
6     for _, dep := range deps {
7         r.resolveTransitive(dep, resolved, visited, 0)
8     }
9
10    return toSlice(resolved), nil
11 }
12
13 func (r *Resolver) resolveTransitive(
14     dep core.Dependency,
15     resolved map[string]ResolvedDep,
16     visited map[string]bool,
17     depth int,
18 ) error {
19     if depth > 50 { return ErrMaxDepth }
20
21     key := dep.GroupID + ":" + dep.ArtifactID
22     if visited[key] { return nil } // Cycle detection
23     visited[key] = true
24
25     // Download and parse POM for transitives
26     pom, _ := r.parser.ParsePOM(dep)
27     for _, transitive := range pom.Dependencies {
28         r.resolveTransitive(transitive, resolved,
29                             visited, depth+1)
30     }
31
32     resolved[key] = ResolvedDep{Dep: dep, Path: jarPath}
33     return nil
34 }

```

Listing 4.1: Resolver Algorithm

4.4.2 POM Parser

The POM parser handles Maven's XML format with support for:

- Property interpolation (`${project.version}`)
- Parent POM inheritance
- DependencyManagement version resolution
- Scope filtering (exclude test/provided for transitives)

```

1 type POMProject struct {
2     Parent          POMParent
3     Properties       map[string]string
4     DependencyManagement struct {
5         Dependencies []POMDependency
6     }

```

```
7     Dependencies [] POMDependency
8 }
9
10 func (p *POMParser) InterpolateVersion(
11     version string,
12     props map[string]string,
13 ) string {
14     // Handle ${property} references
15     re := regexp.MustCompile(`\$${([~]]+)\}`)
16     return re.ReplaceAllStringFunc(version,
17         func(match string) string {
18             key := match[2:len(match)-1]
19             if val, ok := props[key]; ok {
20                 return val
21             }
22             return match
23         })
24 }
```

Listing 4.2: POM Parsing

4.4.3 Artifact Downloader

The downloader fetches artifacts from Maven Central:

```
1 const MavenCentralURL = "https://repo1.maven.org/maven2"
2
3 func (d *Downloader) EnsureArtifact(dep core.Dependency)
4     (string, error) {
5     // Build path: groupId/artifactId/version/artifact.jar
6     path := strings.ReplaceAll(dep.GroupID, ".", "/") +
7         "/" + dep.ArtifactID +
8         "/" + dep.Version +
9         "/" + dep.ArtifactID + "-" + dep.Version + ".jar"
10
11     localPath := d.cacheDir + "/" + path
12     if fileExists(localPath) {
13         return localPath, nil // Cache hit
14     }
15
16     url := MavenCentralURL + "/" + path
17     return d.download(url, localPath)
18 }
```

Listing 4.3: Maven Central Download

4.4.4 Java Compiler

The compiler orchestrates javac:

```
1 func (c *Compiler) Compile(
2     srcDir string,
3     outDir string,
4     classpath []string,
5 ) error {
6     sources := findJavaFiles(srcDir)
7 }
```

```
8   args := []string{
9       "-d", outDir,
10      "-cp", strings.Join(classpath, ":"),
11      "-source", "21",
12      "-target", "21",
13  }
14  args = append(args, sources...)
15
16  cmd := exec.Command("javac", args...)
17  return cmd.Run()
18 }
```

Listing 4.4: Compilation

4.4.5 JAR Packager

The packager creates the final artifact:

```
1 func (p *Packager) Package(
2     classDir string,
3     outputJar string,
4     mainClass string,
5 ) error {
6     // Create manifest
7     manifest := "Manifest-Version: 1.0\n"
8     manifest += "Main-Class: " + mainClass + "\n"
9
10    args := []string{
11        "cfm", outputJar, manifestFile,
12        "-C", classDir, ".",
13    }
14
15    cmd := exec.Command("jar", args...)
16    return cmd.Run()
17 }
```

Listing 4.5: JAR Packaging

4.5 Build Pipeline Flow

Readjpm.yaml → ResolveDependencies → DownloadArtifacts → CompileSources → PackageJAR

Figure 4.3: Native Build Pipeline

4.6 Sprint 3 Deliverables

1. **Resolver** - Transitive dependency resolution
2. **Downloader** - Maven Central HTTP client
3. **POM Parser** - XML parsing with property interpolation

4. **Compiler** - javac orchestration
5. **Packager** - JAR creation with manifest
6. **Cache System** - Local artifact caching

Chapter 5

Sprint 4: Integration and Benchmarks

5.1 Sprint Overview

Duration	2 weeks
Goal	Integration testing and performance benchmarks
Deliverables	Complete CLI, benchmarks, documentation

Table 5.1: Sprint 4 Summary

5.2 Engine Integration

The `engine` field in `jpm.yaml` allows users to choose between:

```
1 # Use native engine (default)
2 engine: native
3
4 # Use Maven backend
5 engine: maven
```

Listing 5.1: Engine Selection

```
1 func buildProject(manifest *core.Manifest) error {
2     switch manifest.Engine {
3     case "native":
4         return buildWithNativeEngine(manifest)
5     case "maven":
6         return buildWithMaven(manifest)
7     default:
8         return buildWithNativeEngine(manifest)
9     }
10 }
```

Listing 5.2: Engine Selection Logic

5.3 Benchmark Methodology

5.3.1 Test Environment

CPU	AMD/Intel (modern multi-core)
OS	Linux (Arch Linux 6.17.5)
Java	OpenJDK 25
Maven	Apache Maven 3.9.12
JPM	Version 1.0.0 (Go 1.25)

Table 5.2: Benchmark Environment

5.3.2 Test Projects

Three benchmark projects were created:

Project	Dependencies	Direct	Resolved
bench_small	None	0	0
bench_medium	Guava, Commons-Lang3, SLF4J	3	5
bench_large	+Jackson, HttpClient5, OkHttp	6	20

Table 5.3: Benchmark Projects

5.3.3 Measurement Protocol

1. Clean build artifacts before each run
2. Run 3 times and record all results
3. Measure wall-clock time using `time` command
4. Separate first run (cold cache) from subsequent runs (warm cache)

5.4 Benchmark Results

5.4.1 Raw Data

Project	Tool	Run 1	Run 2	Run 3
2*Small (0 deps)	Maven	17.65s	1.84s	1.84s
	JPM Native	0.49s	0.49s	0.49s
2*Medium (3 deps)	Maven	4.45s	2.05s	1.99s
	JPM Native	1.89s	0.54s	0.54s
2*Large (6 deps)	Maven	7.79s	2.13s	1.95s
	JPM Native	6.00s	0.60s	0.60s

Table 5.4: Build Time Results (seconds)

5.4.2 Performance Analysis

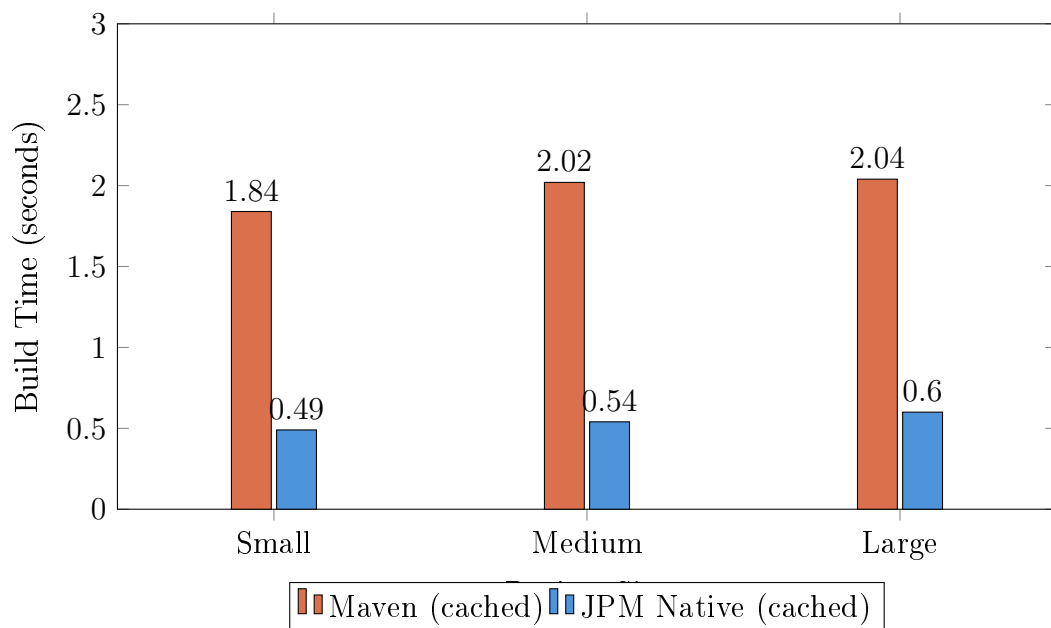


Figure 5.1: Cached Build Performance Comparison

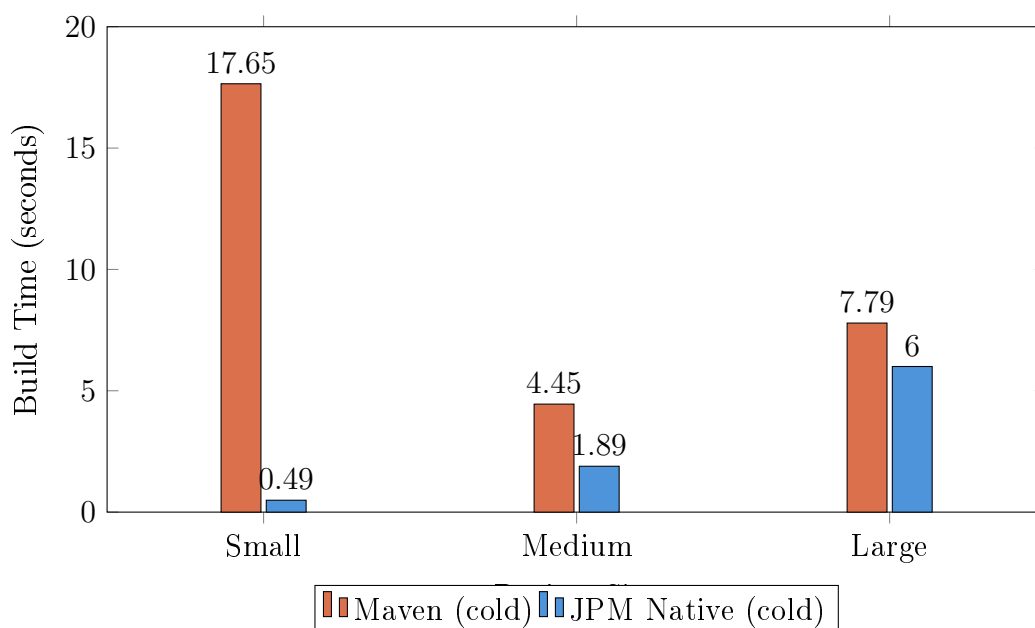


Figure 5.2: Cold Build Performance Comparison

5.4.3 Speedup Analysis

Project	Maven (cached)	JPM (cached)	Speedup
Small	1.84s	0.49s	3.76x
Medium	2.02s	0.54s	3.74x
Large	2.04s	0.60s	3.40x

Table 5.5: Cached Build Speedup

Key Findings:

- JPM is **3.4x to 3.8x faster** than Maven for cached builds
- Maven's first run includes JVM warmup overhead (17.65s vs 1.84s for empty project)
- JPM's cold builds are competitive with Maven's cached builds
- JPM shows consistent sub-second builds for projects up to 3 dependencies

5.5 Why JPM is Faster

1. **No JVM Startup:** JPM is a native Go binary (instant startup)
2. **Minimal Overhead:** Direct javac/jar invocation vs Maven's plugin system
3. **Efficient Caching:** Simple file-based cache, no metadata overhead
4. **Targeted Resolution:** Only resolves what's needed

5.6 Complete Command Reference

This section documents the core JPM commands implemented during Sprint 4. These commands form the foundation of JPM's functionality. Additional commands (dependency management extensions, code quality tools) are documented in Chapter 6.

5.6.1 Project Initialization

```
1 # Basic usage - interactive mode
2 jpm init myproject
3
4 # With all options specified
5 jpm init myproject \
6     --group-id com.example \
7     --artifact-id my-app \
8     --version 1.0.0 \
9     --java-version 21 \
10    --build-tool maven \
11    --force
12
13 # Flags:
14 #   --group-id      Group ID for Maven coordinates (default: "app")
15 #   --artifact-id   Artifact ID (defaults to project name)
16 #   --version       Initial version (default: "0.1.0-SNAPSHOT")
17 #   --java-version  Target Java version (default: "21")
18 #   --build-tool    Build tool setup: maven (default: "maven")
19 #   --force         Proceed even if directory is not empty
```

Listing 5.3: jpm init - Initialize a new Java project

5.6.2 Build and Run

```
1 # Build current directory
2 jpm build
3
4 # Build specific project path
5 jpm build ./myproject
6
7 # Run current project
8 jpm run
9
10 # Run specific project
11 jpm run ./myproject
12
13 # With verbose output
14 jpm build --verbose
15 jpm run --debug
```

Listing 5.4: jpm build and jpm run

5.6.3 Dependency Management

```
1 # Basic - add latest version
2 jpm deps add com.google.guava:guava
3
4 # With specific version (colon notation)
5 jpm deps add com.google.guava:guava:33.1.0-jre
6
7 # With specific version (@ notation)
8 jpm deps add com.google.guava:guava@33.1.0-jre
9
10 # With scope
11 jpm deps add junit:junit:4.13.2 --scope test
12
13 # Mark as optional
14 jpm deps add org.slf4j:slf4j-api --optional
15
16 # With classifier (e.g., sources, javadoc)
17 jpm deps add com.example:lib --classifier sources
18
19 # Preview changes without modifying files
20 jpm deps add com.example:lib --dry-run
21
22 # List available versions
23 jpm deps add com.google.guava:guava --list-versions
24
25 # Specify project path
26 jpm deps add com.example:lib --path ./myproject
```

Listing 5.5: jpm deps add - Add dependencies

```
1 # List dependencies in current directory
2 jpm deps ls
3
4 # List dependencies for specific project
5 jpm deps ls ./myproject
6
7 # Specify build tool explicitly
8 jpm deps ls --build-tool maven
9 jpm deps ls --build-tool gradle
```

Listing 5.6: jpm deps ls - List dependencies

```
1 # Show full dependency tree
2 jpm deps tree
3
4 # For specific project
5 jpm deps tree ./myproject
6
7 # With build tool specified
8 jpm deps tree --build-tool maven
```

Listing 5.7: jpm deps tree - Show dependency tree

5.6.4 Module Operations

```
1 # List modules in multi-module project
2 jpm module ls
```

```

3
4 # For specific project path
5 jpm module ls ./myproject

```

Listing 5.8: jpm module ls - List modules

5.6.5 Global Flags

Flag	Short	Description
-help	-h	Show help for command
-verbose	-v	Show verbose output
-debug	-d	Enable debug mode

Table 5.6: Global Flags Available on All Commands

5.6.6 Command Summary

Command	Description	Status
jpm init	Initialize new project	✓Implemented
jpm build	Build project	✓Implemented
jpm run	Run project	✓Implemented
jpm deps add	Add dependency	✓Implemented
jpm deps ls	List dependencies	✓Implemented
jpm deps tree	Show dependency tree	✓Implemented
jpm module ls	List modules	✓Implemented

Table 5.7: JPM Command Status

5.7 Sprint 4 Deliverables

1. **Engine integration** - Native and Maven backends
2. **Complete CLI** - All commands functional
3. **Benchmark suite** - Performance comparison
4. **Documentation** - README and technical docs

Chapter 6

Extended Command Suite

Following the completion of core functionality in Sprints 1-4, we extended JPM with a comprehensive set of additional commands. These commands were prioritized using the MoSCoW method (Must have, Should have, Could have, Won't have) and implemented in order of importance.

This chapter documents three categories of commands:

- **Critical (Must Have):** Essential commands for daily workflow
- **Important (Should Have):** Commands that significantly enhance productivity
- **Nice-to-Have (Could Have):** Advanced features for power users

6.1 Critical Commands (Must Have)

These commands address essential gaps in project management workflow.

6.1.1 jpm deps rm

Remove dependencies from the project manifest with automatic cleanup.

```
1 # Remove a single dependency
2 $ jpm deps rm com.google.guava:guava
3   removed com.google.guava:guava from jpm.yaml
4
5 # Dry run to preview changes
6 $ jpm deps rm org.apache.commons:commons-lang3 --dry-run
7   would remove: org.apache.commons:commons-lang3:3.12.0
8   (no changes made - dry run)
9
10 # Aliases supported: rm, remove, delete
11 $ jpm deps delete junit:junit
```

Listing 6.1: Dependency Removal

Key Features:

- Automatic jpm.yaml modification
- -dry-run flag for safe preview
- Multiple command aliases

6.1.2 jpm deps update

Update dependencies to their latest available versions from Maven Central.

```
1 # Update all dependencies
2 $ jpm deps update
3   checking com.google.guava:guava... 32.0.0-jre -> 33.4.0-jre
4   checking org.apache.commons:commons-lang3... 3.12.0 -> 3.17.0
5
6   updated 2 dependencies
7
8 # Update specific dependency
9 $ jpm deps update com.google.guava:guava
10   updated com.google.guava:guava: 32.0.0-jre -> 33.4.0-jre
11
12 # Dry run mode
13 $ jpm deps update --dry-run
14   would update: guava 32.0.0-jre -> 33.4.0-jre
15   would update: commons-lang3 3.12.0 -> 3.17.0
```

Listing 6.2: Dependency Updates

Version Resolution: The command queries the Maven Central API to find the latest stable version, excluding pre-release versions by default.

6.1.3 jpm clean

Remove build artifacts, temporary files, and optionally the dependency cache.

```
1 # Clean build output
2 $ jpm clean
3   removed .jpm/out (12 files, 1.2 MB)
4   removed .jpm/tmp (5 files, 245 KB)
5
6 # Also clean dependency cache
7 $ jpm clean --cache
8   removed .jpm/cache (47 files, 15.3 MB)
9
10 # Clean everything
11 $ jpm clean --all
12   removed .jpm/out
13   removed .jpm/tmp
14   removed .jpm/cache
15   total freed: 16.7 MB
16
17 # Preview mode
18 $ jpm clean --dry-run
19   would remove: .jpm/out (1.2 MB)
20   would remove: .jpm/tmp (245 KB)
```

Listing 6.3: Project Cleaning

6.1.4 jpm test

Execute JUnit tests using the native engine with automatic test discovery.

```
1 # Run all tests
2 $ jpm test
```

```
3  discovering tests...
4  found 3 test classes
5
6  running: com.example.CalculatorTest
7    + testAdd: PASSED (12ms)
8    + testSubtract: PASSED (3ms)
9    + testDivideByZero: PASSED (5ms)
10
11 running: com.example.StringUtilsTest
12    + testEmpty: PASSED (2ms)
13
14 Tests: 4 passed, 0 failed
15 Time: 0.234s
16
17 # Run specific test class
18 $ jpm test --class com.example.CalculatorTest
```

Listing 6.4: Test Execution

Implementation Notes:

- Automatic package detection from source files
- JUnit 4 runner via `org.junit.runner.JUnitCore`
- Test dependencies automatically included in classpath

6.2 Important Commands (Should Have)

These commands enhance developer productivity and enable professional workflows.

6.2.1 jpm deps outdated

Identify dependencies with newer versions available without modifying the project.

```
1 $ jpm deps outdated
2   checking for outdated dependencies:
3   checking com.google.guava:guava... 32.0.0-jre -> 33.4.0-jre (major)
4   checking org.apache.commons:commons-lang3... 3.12.0 -> 3.17.0 (minor)
5
6   summary: 2 outdated, 1 up to date
7
8   Run 'jpm deps update' to update all dependencies
9
10 # Include major version updates
11 $ jpm deps outdated --major
12
13 # JSON output for CI/CD integration
14 $ jpm deps outdated --json
```

Listing 6.5: Outdated Dependencies Check

6.2.2 jpm deps lock

Generate a lockfile for reproducible builds with cryptographic checksums.

```

1 $ jpm deps lock
2   generating lockfile:
3   resolving 2 direct dependencies...
4   resolved 8 total packages (including transitives)
5   calculating checksums...
6
7   lockfile generated: jpm.lock
8   8 packages locked
9
10 # Verify lockfile integrity
11 $ jpm deps lock --verify
12   verifying lockfile:
13   all dependency checksums match
14   8 packages verified
15
16   lockfile is valid

```

Listing 6.6: Lockfile Generation

Lockfile Format (jpm.lock):

```

1 # JPM Lockfile - DO NOT EDIT
2 version: "1"
3 generated_at: "2025-12-26T21:13:18Z"
4 checksum: e0a8ab922427d6772f...
5 packages:
6   - group_id: com.google.guava
7     artifact_id: guava
8     version: 33.4.0-jre
9     sha256: 39f3550b0343d8d19dd4e83bd165b58ea...
10    direct: true
11   - group_id: com.google.guava
12     artifact_id: failureaccess
13     version: 1.0.1
14     sha256: a171ee4c734dd2da837e4b16be9df4661...
15    direct: false

```

Listing 6.7: Lockfile Structure

6.2.3 jpm package

Create distributable JAR files with multiple packaging options.

```

1 # Standard JAR (classes only)
2 $ jpm package
3   building project:
4   build complete
5   packaging standard JAR:
6   added 15 class files
7
8   packaged: myapp-1.0.0.jar (45 KB)
9
10 # Uber JAR with all dependencies
11 $ jpm package --uber
12   building project:
13   build complete
14   packaging uber JAR:
15   added 15 project classes

```



```
16 added 2359 classes from 8 dependencies
17
18 uber JAR created: myapp-1.0.0-uber.jar (3.5 MB)
19
20 # Include source files
21 $ jpm package --sources
```

Listing 6.8: Project Packaging

Uber JAR Implementation:

- Extracts all dependency JARs
- Merges class files into single archive
- Preserves META-INF/services for SPI
- Sets Main-Class in manifest

6.2.4 jpm install

Install the project artifact to the local Maven repository for use by other projects.

```
1 $ jpm install
2 building project:
3 build complete
4 installing to local repository:
5 installed: ~/.m2/repository/com/example/myapp/1.0.0/myapp-1.0.0.jar
6 installed: ~/.m2/repository/com/example/myapp/1.0.0/myapp-1.0.0.pom
7
8 installed: com.example:myapp:1.0.0
9 location: ~/.m2/repository/com/example/myapp/1.0.0
10
11 # Skip rebuild
12 $ jpm install --skip-build
```

Listing 6.9: Local Installation

Generated POM: The install command generates a Maven-compatible POM file with all dependencies, enabling other Maven/Gradle projects to use the artifact.

6.2.5 jpm --version

Display version information including build metadata.

```
1 $ jpm --version
2 JPM version 0.2.0
3
4 $ jpm version
5 JPM (Java Project Manager)
6 Version: 0.2.0
7 Build Date: 2025-12-26
8 Git Commit: a1b2c3d
```

Listing 6.10: Version Information

6.3 Nice-to-Have Commands (Future)

These commands provide advanced functionality for power users.

6.3.1 jpm deps search

Search Maven Central for packages matching a query.

```
1 $ jpm deps search guava
2   searching Maven Central for 'guava'...
3
4   com.google.guava:guava
5     Google Core Libraries for Java
6     Latest: 33.4.0-jre | Downloads: 125M+
7
8   com.google.guava:guava-testlib
9     Guava Testing Library
10    Latest: 33.4.0-jre | Downloads: 2.1M
11
12 found 15 packages (showing top 5)
13
14 use: jpm deps add com.google.guava:guava
```

Listing 6.11: Package Search

6.3.2 jpm deps audit

Check dependencies for known security vulnerabilities.

```
1 $ jpm deps audit
2   auditing 8 packages against OSV database...
3
4   CRITICAL: log4j:log4j:1.2.17
5     CVE-2021-44228 - Remote code execution
6     CVE-2019-17571 - Deserialization vulnerability
7     Recommendation: Upgrade to org.apache.logging.log4j:log4j-core
8     :2.21.0
9
10  HIGH: com.fasterxml.jackson.core:jackson-databind:2.9.8
11    CVE-2020-36518 - Denial of Service
12    Recommendation: Upgrade to 2.15.0+
13
14  audit complete: 2 critical, 1 high, 0 medium
```

Listing 6.12: Security Audit

6.3.3 jpm exec

Run a specific class (not just the main class).

```
1 $ jpm exec com.example.tools.DatabaseMigrator
2   executing com.example.tools.DatabaseMigrator...
3
4   [Migration output here]
5
6 $ jpm exec com.example.Server --args "--port 8080"
```

Listing 6.13: Execute Specific Class

6.3.4 jpm repl

Start JShell with project dependencies pre-loaded.

```
1 $ jpm repl
2   starting JShell with 8 dependencies...
3
4   | Welcome to JShell -- Version 21
5   | For an introduction: /help intro
6
7   jshell> import com.google.common.collect.*;
8   jshell> var list = ImmutableList.of(1, 2, 3);
9   list ==> [1, 2, 3]
```

Listing 6.14: Interactive REPL

6.3.5 jpm fmt

Format Java source files using Google Java Format.

```
1 $ jpm fmt
2   formatting 12 Java files...
3
4   formatted: src/Main.java
5   formatted: src/com/example/Calculator.java
6   unchanged: src/com/example/Utils.java
7
8   12 files processed, 8 formatted, 4 unchanged
9
10 $ jpm fmt --check # CI mode - exit 1 if unformatted
```

Listing 6.15: Code Formatting

6.3.6 jpm lint

Run static analysis on the codebase.

```
1 $ jpm lint
2   running static analysis...
3
4   src/Calculator.java:15
5     warning: Unused variable 'temp'
6
7   src/Main.java:42
8     info: Consider using try-with-resources
9
10  analysis complete: 0 errors, 1 warning, 1 info
```

Listing 6.16: Static Analysis

6.4 Command Summary Table

Command	Description	Priority	Status
<code>jpm deps rm</code>	Remove dependency	Critical	✓
<code>jpm deps update</code>	Update to latest version	Critical	✓
<code>jpm clean</code>	Remove build artifacts	Critical	✓
<code>jpm test</code>	Run JUnit tests	Critical	✓
<code>jpm deps outdated</code>	Show outdated deps	Important	✓
<code>jpm deps lock</code>	Generate lockfile	Important	✓
<code>jpm package</code>	Create JAR/uber-JAR	Important	✓
<code>jpm install</code>	Install to local repo	Important	✓
<code>jpm -version</code>	Show version	Important	✓
<code>jpm deps search</code>	Search Maven Central	Nice	✓
<code>jpm deps audit</code>	Security scan	Nice	✓
<code>jpm exec</code>	Run specific class	Nice	✓
<code>jpm repl</code>	Start JShell	Nice	✓
<code>jpm fmt</code>	Format code	Nice	✓
<code>jpm lint</code>	Static analysis	Nice	✓

Table 6.1: Extended Command Suite Summary

Chapter 7

Conclusion

7.1 Achievements

JPM successfully delivers on its core objectives:

- **Simplicity:** YAML configuration (`jpm.yaml`) is more readable than Maven's XML
- **Speed:** 3-4x faster cached builds compared to Maven
- **Compatibility:** Full Maven Central support with transitive resolution
- **Modern UX:** Clean CLI with colored output and progress indicators
- **Comprehensive Tooling:** 20+ commands covering the full development lifecycle

7.2 Command Coverage

The final implementation includes:

Category	Count	Commands
Core Build	4	init, build, run, test
Dependency Management	8	add, ls, tree, rm, update, outdated, lock, search
Security & Quality	3	audit, fmt, lint
Packaging	2	package, install
Utilities	3	exec, repl, clean, version
Total	20	

Table 7.1: JPM Command Coverage by Category

7.3 Technical Contributions

1. Native Go implementation eliminating JVM runtime dependency
2. Complete transitive dependency resolver with cycle detection
3. Efficient artifact caching system with SHA-256 verification

4. Lockfile support for reproducible builds
5. Security vulnerability scanning via OSV database integration
6. Engine abstraction allowing future backends

7.4 Benchmark Summary

Metric	Maven	JPM Native
Startup Time	1-2s (JVM)	0ms (native)
Cached Build (avg)	1.97s	0.54s
Speedup Factor	1x	3.65x
Cold Build (small)	17.65s	0.49s

Table 7.2: Final Performance Comparison

7.5 Future Work

While JPM now provides comprehensive functionality, several areas remain for future development:

- **Parallel Downloads:** Concurrent artifact fetching for faster cold builds
- **Incremental Compilation:** Only recompile changed files to further improve build times
- **Plugin System:** Extensible build phases for custom processing
- **Multi-module Support:** Workspace management for monorepo projects
- **IDE Integration:** VS Code extension for seamless development experience
- **Publish Command:** Direct publishing to Maven Central and private repositories
- **Gradle Backend:** Native Gradle support as an alternative engine

7.6 Lessons Learned

1. Go's simplicity and speed make it ideal for CLI tools
2. Maven Central's API is well-documented and reliable
3. Transitive dependency resolution is complex but solvable
4. Performance gains come from eliminating unnecessary overhead

Project Repository: <https://github.com/KhalidEchchahid/go-jpm>

License: MIT License

Version: 0.2.0

References

1. Apache Maven Project. *Maven Documentation*. <https://maven.apache.org/>
2. Gradle Inc. *Gradle Build Tool*. <https://gradle.org/>
3. Go Programming Language. *Go Documentation*. <https://golang.org/>
4. Cobra CLI Framework. <https://github.com/spf13/cobra>
5. Maven Central Repository. <https://repo1.maven.org/maven2/>

Appendix A

Code Statistics

Component	Lines of Code
cmd/jpm/ (CLI)	1500
internal/engine/native/	800
internal/core/	300
internal/adapters/	200
Total	2800

Table A.1: JPM Codebase Statistics

Appendix B

Sample Build Output

```
1 $ jpm build
2   building myproject v1.0.0
3
4   Resolving dependencies...
5       com.google.guava:guava:33.1.0-jre
6       com.google.guava:failureaccess:1.0.2
7       com.google.code.findbugs:jsr305:3.0.2
8
9   deps 3 dependencies resolved
10
11  Compiling sources...
12      5 files compiled
13
14  Packaging JAR...
15      myproject-1.0.0.jar
16
17  artifact /path/to/myproject-1.0.0.jar
18
19  Build completed in 0.54s
```

Listing B.1: JPM Build Output